Mobile Quad-Controlled Wireless Robotic Arm

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***Abstract*—In this paper, we develop a remotely controlled robotic arm with 4 degree of freedom (D.O.F) that is wirelessly controlled using four control mechanisms, i.e, Voice Control, Smart Phone-Tilt Control, Remote control and Hand Gesture Control. Wireless technologies such as Bluetooth and Wi-Fi are used to access the Quad-Controlled Robotic Arm (QCRA).**

A prototype QCRA is developed. The QCRA can be used to pick and place objects from one place to another on receiving the commands from distance, thereby reducing the human effort. An Android application is developed for convenience of the user in operating this QCRA, using different control mechanisms. Performance evaluation results are encouraging. Potential applications of the QCRA in homes, industries and for physically challenged/aged people are also discussed.

Keywords—Quad-controlled Robotic Arm; Cloud-Server; Wi-Fi; Bluetooth; Hand-gesture Recognition; Voice-control; Smart Phone-tilt Control; Remote Control

I. INTRODUCTION

In today's sophisticated world; robots and autonomous machines are an integral part of daily routine. They help in completing daily chores for human beings; especially for physically challenged and aged people of the society. These robots are also critical in many industries as they reduce life casualties and financial degradation while performing risky and critical operations. Over the last few years; there has been a huge amount of work done in this area; Voice [1] and Speech Control Robots have been developed [2]. In Brown University; a robotic arm was designed that could be controlled by human brain [3]. A robot was used to drill boreholes in con-gested urban subsurface areas [4]. Robots are also developed using artificial intelligence [5] and PID (Proportional-Integral-Derivative) control [6] method-logies.

There has also been a lot of development in the area of image processing in the past few years [7]. Image pro-cessing techniques have been applied increasingly for good quality evaluation in recent years. Recent advances in image processing techniques include charge coupled device cam-era; ultrasound; magnetic resonance imaging; computed to-mography; and electrical tomography for image acquisition; pixel and local pre-processing approaches for image pre-processing; thresholding-based; gradient-based; region-based; and classification-based methods for image segmentation; size; shape; colour; and texture features for object measurement; and statistical; fuzzy logic; face detection and recognition [8]; neural network methods for classification [9].

Hand Gesture Recognition [10]; based on image processing is the ability to track a person's hand movements and de-termine what gestures they may be performing which can be achieved through various tools. In this paper; we use MATLAB

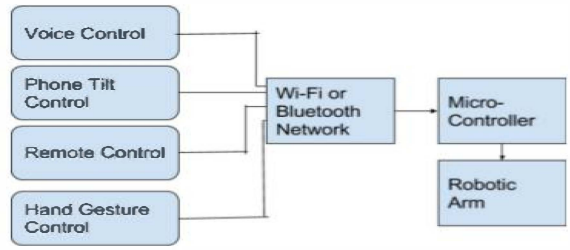

Fig. 1: Schematic Control Diagram of the Quad-Control Robotic Arm

which is a high-performance language for technical comput-ing. First; Image acquisition is done via the android application in a Smart Phone; then the Hand gesture is recognized by pattern matching using PCA (Principal component analysis) algorithm [11] and the respective instruction corresponding to the matched gesture is transmitted to the robot; to perform a specified action.

Since the advent of Android applications by Android; Inc.; which Google bought in 2005; controlling robotic devices through Smart Phones remotely has become simple and ef-fortless.These applications which extend the functionality of devices; are written using the Android software development kit (SDK) and; often; the Java programming language that has complete access to the Android APIs. In this paper; we display an android application that is created by us using MIT-App inventor which is a cloud-based tool for android application development [12]. The application provides an interface for controlling the QCRA via the illustrated mechanisms; i.e.; Smart Phone Tilt [13]; Voice [14]; Hand Gesture [15] and Remote [16] using either of the wireless networks-Bluetooth or Wi-Fi.

Multiple control mechanisms[17] increase reliability and efficiency of the device as one mechanism can compensate the limitations of the other. For example; Wi-Fi can be used in cases where the operation range must be high and Bluetooth Fails.

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This paper is organized as follows: In Section 2; wireless operation details and control mechanisms used are discussed. Section 3 discusses the experiments conducted to measure the efficiency of operation and observations made. In Section 4 the results are discussed and inferences are made. A summary of the paper is given in Section 5; along with scope of further work.

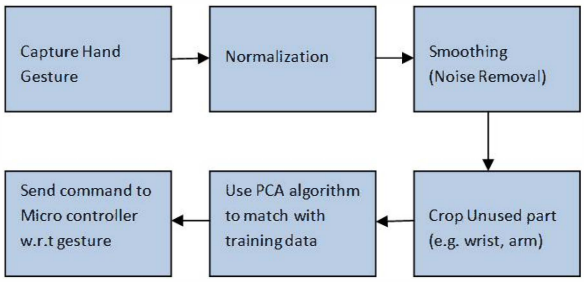

Fig. 2: Hand-Gesture Recognition Control Flow Diagram

II. Wireless Operation of the QCRA and Control Mechanisms Used

A. *Wireless Communication Networks*

The wireless operation of the QCRA is done through two standards of communication which are Wi-Fi and bluetooth. These two technologies increase the reliability as each of them can compensate for the others constraints. Bluetooth is preferred when transferring information between two or more devices that are near each other without the requirement of a common network. However; Wi-Fi is better suited for operat-ing full-scale networks because it enables a faster connection; better range from the base station; and better wireless security.

In the QCRA that we have developed we use the ESP8266 Wi-Fi module [18] . It is a self contained System-On-Chip(SOC) with integrated TCP/IP protocol stack that can give any micro -controller access to your WiFi network. The ESP8266 is capable of either hosting an application or offload-ing all Wi-Fi networking functions from another application processor. Each ESP8266 module comes pre-programmed with an AT command set firmware; which means; it can be simply hooked up to the micro-controller device and get about as much Wi-Fi ability as a Wi-Fi Shield offers. This module has a powerful enough on-board processing and storage capability that allows it to be integrated with the sensors and other application specific devices through its GIPO(General-purpose input/output) with minimal development up-front and minimal loading during run-time. The ESP8266 supports Automatic Power Save Delivery(APSD) for VoIP applications and Blue-tooth co-existance interfaces; it contains a self-calibrated RF allowing it to work under all operating conditions; and requires no external RF parts.

The other communication technology used is Bluetooth; which is a wireless technology standard for exchanging data over short distances. We use the HC-05 module which is an easy to use Bluetooth SPP (Serial Port Protocol) module; designed for transparent wireless serial connection setup.

Serial port Bluetooth module is fully qualified Bluetooth V2.0+EDR (Enhanced Data Rate) 3Mbps Modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR Bluecore 04-External single chip Bluetooth system with CMOS technology and with AFH(Adaptive Frequency Hop-ping Feature).

B. *Control Mechanisms*

a. *Voice Control*: Voice Control is achieved by incor-porating the Google web speech API [19] in the android application; which enables speech recognition of the user and synthesis. The Voice commands illustrated in the below table are used to control the QCRA. These commands are converted to corresponding text and instructions are transmitted to the micro-controller; via Bluetooth or Wi-Fi; which results in the movement of the QCRA.

b. *Phone Tilt Control*: The phone tilt control works on the change in orientation of the phone which can be measured by the inbuilt accelerometer in the phone; the change of axis detected by the accelerometer will be transmitted via Blue tooth or Wi-Fi to the micro-controller and the corresponding movement with respect to the recieved axis information.

c. Remote Control: A simple interface is made in the android application to transfer commands to the QCRA; using touch buttons. The command can be transmitted to the micro-controller via Wi-Fi or Bluetooth. Figure 5(b) depicts the user interface of the application for remote control.

d. *Hand Gesture Control*: Hand Gesture Recognition is incorporated by using MATLAB language integrated with Arduino IDE. The Principal component Analysis algorithm is used to achieve this task. Initially an image is captured via the camera of the Smart Phone; as shown in Figure 5(a). It is assumed that while capturing the image; black background with proper light source is used. Sometimes it is possible that the Image; with a gesture contains extra part along with required part of hand i.e. background objects; blank spaces etc. For better results in the pattern matching procedure; the unused part must be removed. Therefore; hand gesture is cropped out from the obtained frame. This cropping happens in 3 steps. First; the image must be converted into HSV(Hue; Saturation; Value) color space [20] . A HSV based skin filter must be applied on the RGB format image to reduce lighting effects. Finally; the resultant image must be segmented to get a binary

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image of hand. Binary images are bi-level images where each pixel is stored as a single bit (0 or 1). Smoothening is required; to reduce the noise in the input image. After the binary image has been created; it is compared with the images in the training database and the corresponding gesture number is displayed. The respective instruction is executed. Table 2 shows the list of gestures used and respective commands while figure 6 depicts the gestures used.

III. Experiments and Observations

Certain experiments were performed on the QCRA in order to increase the efficiency as well as to scale the performance. They are as follows:

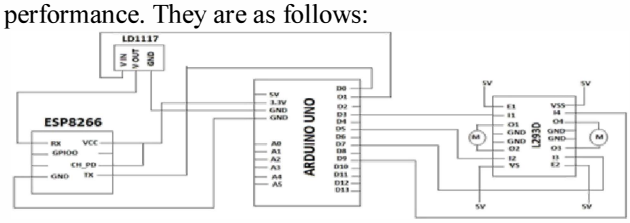

Fig. 3: ESP8266 Connection to Arduino [18]

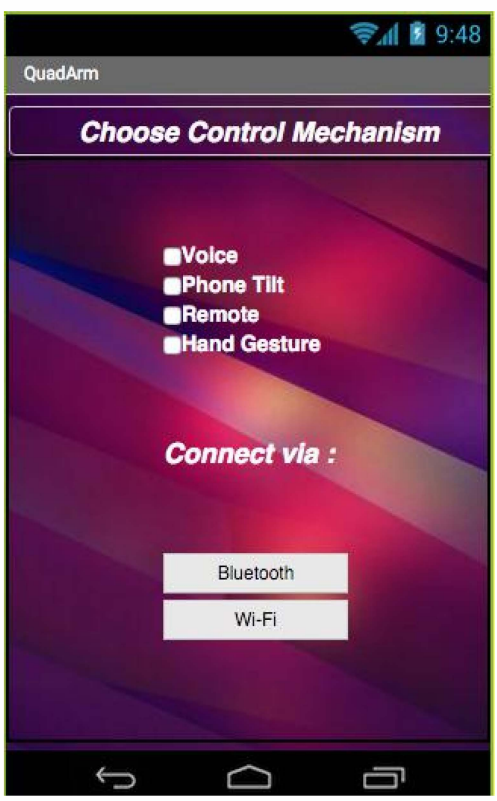

Fig. 4: Home Page-Android based Application

A. *Experiment 1: Study of Change in Response Time of the QCRA in Voice Control with different Languages*

To scale the performance of the QCRA; we chose a few languages which are among the most spoken languages in the world; excluding English; i.e.; Hindi; Bengali; Russian; Spanish and French; and implemented voice control with them. We used the Google Translate API [21] in our Android application which can translate voice inputs of these languages to text. The translated texts can be further processed to transmit instructions. Table 4 depicts the response time for these languages for commands transmitted via Bluetooth. We calculate the time overhead that is consumed for translation. This value will be 0.00 for English as it is not translated. We assume that the Bluetooth latency is fixed.

TABLE 1: HAND GESTURES WITH ITS CORRESPONDING MOVEMENT

(a) Gesture Depicting Numbers	(b) Movement
One	QCRA's body moves Up
Two	QCRA's body moves Down
Three	QCRA's Body moves Forward
Four	QCRA's Body moves Backward
Five	QCRA's Body turns towards Right
Six	QCRA's Body turns towards Left
Seven	QCRA's body turns Clockwise
Eight	QCRA's body turns Anti-Clockwise
Nine	QCRA's claw opens up
Ten	QCRA's claw gets closed
Closed Fist	QCRA Stops

TABLE 2: STUDY OF RESPONSE TIME OF QCRA WITH DIFFERENT LANGUAGES

(a) S. No.	(b) Language	(c) Response Time (in sec)	(d) Time Overhead (in sec)
1	English	3.613	0.00
2	Hindi	3.634	0.021
3	Bengali	3.946	0.333
4	French	3.825	0.212
5	Spanish	3.998	0.385
6	Russian	4.345	0.732

B. *Experiment 2: Comparison of Time Latency Between Wi-Fi and Bluetooth for Remote Control*

Wi-Fi technology is advantageous as compared to Bluetooth as it can operate through larger distances and time latency is very low. However; Wi-Fi requires a network for operation. In this experiment; we draw a comparison between the time latency of Wi-Fi and Bluetooth communication for the Smart Phone remote control mechanism. We consider several ranges of operation. Table 4 depicts the experiment and Figure 7 gives a graphical representation.

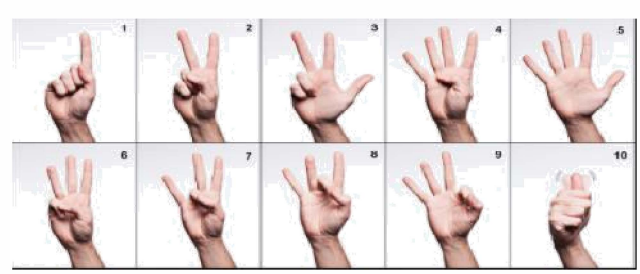

Fig. 5: Hand Gestures that are used as Commands/Instructions

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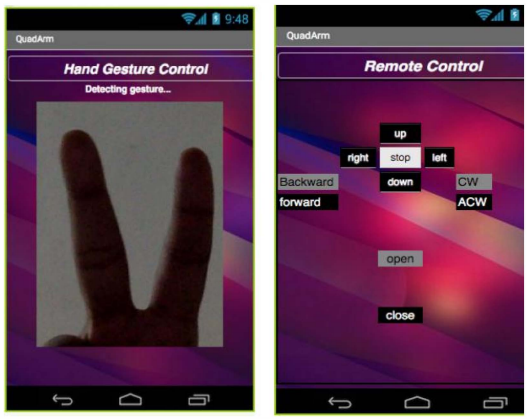

(a) Hand Gesture Control (b) Remote Control

Fig. 6: User Interface of the Android Based Application

TABLE 3: VOICE COMMANDS CONTROLLED WITH DIFFERENT MOVEMENTS

(a) Voice Command	(b) Movement
Up	QCRA's body moves Up
Down	QCRA's body moves Down
Forward	QCRA's Body moves Forward
Backward	QCRA's Body moves Backward
Right	QCRA's Body turns towards Right
Left	QCRA's Body turns towards Left
Anti-Clockwise	QCRA's body turns Anti-Clockwise
Anti-Clockwise	QCRA's body turns Anti-Clockwise
open	QCRA's claw opens up
close	QCRA's claw gets closed
Stop	QCRA Stops

C. *Experiment 3 : Study of Gesture Recognition Rate*

The Hand gesture recognition methodology used converts the image into a binary image and implements a pattern matching algorithm; i.e.; computing the Eigen distances to find the closest matching gesture. Since human gestures can be fuzzy or arbitrary sometimes; recognition can go wrong. In this experiment; we try recognizing each gesture number of times to find the efficiency with which the gesture gets correctly detected. Table 5 depicts the experiment.

IV. RESULTS AND DISCUSSION

The QCRA has certain limitations and specifications. The payload (the maximum weight that the QCRA's claw can hold without any De-shaping) is restricted to about 420 grams; whereas the motor can bear a torque of 1.25 kg. Therefore the maximum weight it can lift theoretically is 1.25kg. However; the practical maximum possible weight it lifts is only about 420 grams. If the payload is more than 420 grams; then the shape of the claws is altered. It means that the maximum payload is dependent on the strength of the QCRAs claws.

As per the observations made in the previous section we noticed that the maximum range of Bluetooth is approximately 9m to 11m.In the First Experiment; from the time latency measures for various languages; it was observed that the latency time was maximum for the Russian Language . We drew a comparison between Bluetooth and Wi-Fi time latencies for various ranges; in Experiment 2. Since Bluetooth operates over smaller ranges; after about 11m the signals are not received and hence instructions do not get executed. The signals from Wi-Fi start diminishing at a distance of about about 470m. In the last experiment we find the efficiency of our Hand-Gesture Recognition System by computing the Recognition rate of each gesture. The Highest rate was observed for gesture 3.

V. SUMMARY AND CONCLUSIONS

In this paper; an interactive Quad Control Robotic Arm which can work on four different control mechanisms and two different communication protocols; was developed. This QCRA can be used as an assistant for picking and placing objects at far-off distances for physically challenged and aged people [22]. Its applications can be found in Industries [23] and hospitals [24] also. Operation and functioning of all these mechanisms were discussed in short. A brief study is carried out analyzing the performance of the QCRA. Time latencies are calculated at various ranges of operation/distances for the Communication networks used; i.e; Bluetooth and Wi-Fi. The efficiency of the Hand-Gesture Recognition System is found by conducting a number of trails on all the gestures.

TABLE 4: COMPARISON BETWEEN BLUETOOTH AND WI-FI TIME LATENCIES ACCORDING TO THE DISTANCE BETWEEN USER AND QCRA

(a) S. No.	(b) Distance between User and QCRA (in m)	(c) Time la-tency (Bluetooth) (in sec)	(d) latency(Wi- Fi) (in sec)	Time
1	3.00	1.168	0.281	
2	4.50	1.721	0.447	
3	6.00	2.432	0.735	
4	7.50	2.994	0.970	
5	9.50	3.091	1.116	
6	11.00	Signal Not Re-	1.324	
		ceived		
7	>479.00	Signal Not Re-	Signal	Not
		ceived	Recieved	

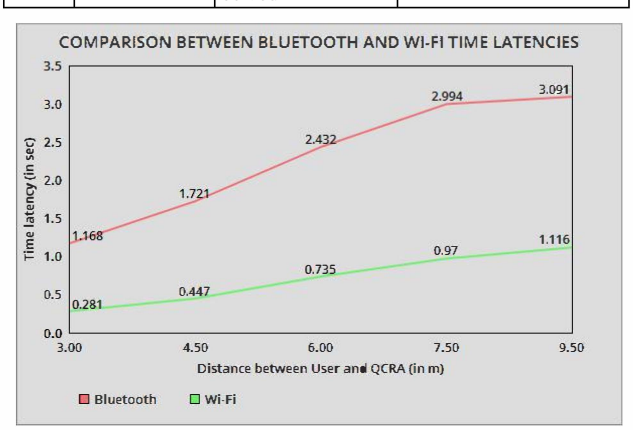

Fig. 7: Hand Gestures that are used as Commands/Instructions by QCRA

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Areas that can be explored further are; the effect of the distance between the camera of the Smart-Phone and the user's hand on the performance of the robot; the effect of noise on the speech to text conversion; Decreasing the time latency for Hand Gesture Recognition. The cost of the QCRA can possibly be reduced by using renewable sources of energy. The weight carrying capacity of the robot can be increased by using a stronger metal of the arms; and the motors with higher torque.

TABLE 5: GESTURE RECOGNITION RATE OF NUMBERS WITH ITS CORRESPONDING NUMBER OF TRAILS

(a) S. No.	(b) Gesture Depicting Number	(c) Number of Trials	(d) Gestures Correctly Identified	(e) Recognition Rate (in %)
1	1	10	9	90
2	2	10	7	70
3	3	10	10	100
4	4	10	8	80
5	5	10	9	80
6	6	10	8	80
7	7	10	7	70
8	8	10	8	80
9	9	10	9	90
10	10	10	9	90
11	0	10	7	70

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